Day 1: Tools for Troubleshooting

Computer Hardware Basics • Assignment

Objective: To learn how a multimeter, POST card, and software diagnostic tools help identify hardware problems.

Task 1: Three Uses of a Multimeter

Situation	How the multimeter helps
Laptop charger is not working	Measures the charger's output voltage to check whether it is supplying power correctly.
Phone is not charging	Can check voltage or continuity in an accessible cable/connector; testing should be done safely.
Desktop PC will not turn on	A technician can measure power-supply outputs to help find a faulty PSU or connection.

Task 2: POST Card Video Summary

A POST card is used when a computer does not start normally. During startup, it displays diagnostic codes generated during the motherboard's Power-On Self-Test (POST). A technician checks the code against the motherboard or POST-card manual to identify the stage where startup stopped. Depending on the code, the issue may involve memory, CPU initialization, or another hardware component.

Video to watch: YouTube search: POST card motherboard test. The exact code and fault depend on the video selected, so add the specific code shown in your chosen demo before submitting.

Task 3: Software Diagnostic Tool

Tool selected: HWMonitor. It can display available hardware readings such as CPU/GPU temperatures, voltages, fan speeds, and drive temperatures. The readings shown depend on the computer's hardware.

Official download: CPUID HWMonitor download page.

Steps to complete the practical:

1. Open the official download page and download HWMonitor.
2. Install it, then open the program.
3. Check the sensor readings on the main screen.
4. Take a screenshot using **Windows + Shift + S** and save it.
5. Crop or blur any personal information before adding it to your GitHub submission.

Screenshot: Add your own HWMonitor screenshot here. It must come from the program running on your computer; a sample image would not prove that the practical was completed.

Task 4: Comparison of Troubleshooting Tools

Tool	What it checks	Example problem detected
Multimeter	Voltage, resistance, and continuity in electrical circuits.	Charger or power supply giving incorrect voltage.
POST card	Motherboard startup/POST diagnostic codes.	Startup stops during a hardware initialization stage.
Software diagnostics	Available temperatures, sensor readings, and drive-health data.	High CPU temperature or a drive-health warning.

Conclusion: A multimeter checks electrical values, a POST card helps diagnose startup errors, and software tools show system sensor information. Choosing the right tool makes troubleshooting easier and more accurate.